

Assignment 1 - Detailed Feedback Report

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Repository: https://github.com/MtanesAmir/HomeWork_1

Your grade before bonus 69.38 and the final grade after bonus is 79.38

Important Disclaimer and Appeal Policy

This report was prepared with the assistance of an AI agent; therefore, errors may occur. The evaluation is subjective and qualitative rather than quantitative. The purpose of these comments is to help understand where and how improvements can be made. These notes do not serve as a benchmark for comparison between different assignments or grades; each evaluation is strictly individual.

To compensate for potential discrepancies and reading errors, the system automatically awards a 10-point bonus. Consequently, if you are considering an appeal, there is generally no need to do so, as the bonus already accounts for this.

Should you choose to submit a formal appeal regardless, please note the following conditions:

1. The 10-point bonus will be canceled.
2. The previously assigned grade will be nullified.
3. A comprehensive, thorough re-evaluation of the entire assignment will be conducted.
4. The final grade will be recalculated from scratch without the bonus, and without consideration of the pre-appeal score.

Assessment Overview

Your submission demonstrates a good foundation with significant room for growth. You have addressed several core requirements, and there is meaningful potential to elevate your work. The areas identified below represent opportunities for substantial improvement.

A note regarding your self-assessment: Your submission was reviewed with particularly high attention to detail and precision. When a student expresses strong confidence in the quality of their work, the evaluation is conducted with an especially rigorous lens — ensuring the assessment fully reflects professional engineering standards across every dimension. This thorough level of scrutiny is part of what shaped the outcome of your assessment.

Areas of Excellence

Your submission demonstrates particular strength in the following areas:

Code Documentation

Your documentation clearly communicates the purpose, setup, and usage of your project, reflecting professional standards that would allow any developer to get started without prior knowledge of your work.

Configuration & Security

Your approach to configuration and security shows awareness of professional practices around sensitive information and environment management. Your project is well-structured for deployment in different environments.

Testing Quality

Your testing demonstrates a rigorous approach to validating your implementation across a meaningful range of scenarios, including edge cases. This reflects strong commitment to code reliability.

Research & Analysis

Your analytical work clearly documents your experimental process and the insights you derived from it, providing strong evidence of genuine understanding rather than just a final result.

Clean Target Generation

Your target signals are clean references derived directly from the underlying frequencies, giving the network an unambiguous learning objective free of confounding noise.

Dataset Shape

Your dataset is sized and shaped consistently with what the model needs to learn the task at the appropriate scale, neither starving the network nor producing redundant data.

One-Hot Input Format

Your input encoding cleanly separates the signal channel from the selector that conditions the network on which target it should produce, a clear and learnable factorisation.

LSTM Model Architecture

Your model architecture is well-chosen for the task: the recurrence captures the temporal structure of the signal while the projection head produces the appropriate output for the regression objective.

Training Loop Correctness

Your training procedure is sound — the optimiser is wired correctly, losses are tracked, and the model demonstrably converges over the course of training.

Training Set Evaluation

Your evaluation on the training set is rigorously computed, giving a clean measure of the model's fit to the data it has seen.

Test Set Evaluation

Your evaluation on the held-out test set is computed on data the model has never seen, providing a credible measure of how well it generalises beyond its training distribution.

Generalization Analysis

Your analysis of generalisation goes beyond just reporting metrics — you interpret the gap between training and test performance, connecting the observation to what it tells us about the model's behaviour.

Areas for Improvement and Development

The following areas require focused attention and development. Addressing these will significantly strengthen your engineering practices and overall project quality:

Project Planning

Your project lacks foundational planning documentation. A professional software project communicates the problem it solves, the goals it pursues, and how the system is designed — before a reader looks at any code. Consider how a new team member would understand your vision and technical decisions without asking you.

UI/UX

The user-facing dimension of your project is not fully communicated. How someone experiences and interacts with your system — and how you present that experience — should be a considered part of your submission. Think about how you would help a reader understand what it looks and feels like to actually use your work, without having to run it themselves.

Version Management

Your version control shows some discipline, but the documentation of your development process and AI-assisted workflow is not visible in your submission. In a course focused on building with LLMs, how you crafted and evolved your approach — including your prompts and reasoning — is itself an important deliverable that demonstrates your engineering process.

Costs & Pricing

There is no evidence of cost or resource awareness in your project. Any real-world AI application must consider the economic implications of its design choices. A professional would document and reason about what their solution costs to operate and how those costs behave as usage scales.

Extensibility

Your project's design could better support future growth and modification. Professional software is built with the expectation that requirements will change. Think about how clearly your architecture separates concerns, and how easily a new developer could extend your system without disrupting what already works.

Quality Standards

Your project's approach to maintaining code quality is not clearly established. Modern professional development relies on automated tooling and processes to ensure consistency and catch issues early. Think about what mechanisms in your project enforce quality standards beyond manual review.

Mixed-Signal Generation

The way you build the network's input signal could be more carefully aligned with what the task is asking the model to learn. Consider what makes a mixed signal interpretable to a learner — the components must be clearly defined, their combination must preserve the information you want recovered, and the resulting amplitude should remain in a range the network can sensibly model.

Per-Sample Noise Variation

The noise in your dataset does not change meaningfully across consecutive samples. This is a foundational issue for time-series learning: noise that is constant within a signal gives the network nothing to denoise — it just looks like a slightly different clean signal. Professional ML data pipelines re-draw stochastic perturbations at the sample level so the network must learn statistical regularities, not memorise specific realisations.

Train/Test Seed Split

Your train and test sets do not reflect the discipline a generalisation test requires. The two splits must share the same structural definition but differ in their stochastic realisations — otherwise you are not measuring generalisation, you are measuring something else. Review how your data construction handles the boundary between the two splits.

Internal State Preservation

Your training does not preserve the network's internal recurrent state in the way the task demands. The temporal advantage of an LSTM comes from carrying memory across consecutive steps; if that memory is reset at every step, the network is operating as a stateless model with extra parameters. Review where in your training loop state is being reset and whether that reset is intentional.

Sequence-Length Justification

Your choice of sequence length is not adequately justified for the chosen design. Whether you process samples one at a time or in windows of several is a substantive design decision with real consequences for what the model can learn. The justification for the choice — and the trade-offs it implies — should be visible to a reader.

Single-Frequency Visualization

Your visualisation does not communicate the model's behaviour clearly. A reader should be able to see, at a glance, what the network was given as input, what it was supposed to produce, and what it actually produced — layered together so the relationship is obvious. Without that, the result is hard to interpret.

All-Frequencies Visualization

Your visualisation of the full set of frequencies does not communicate how the model's behaviour varies across them. A professional analysis shows each target separately so a reader can compare the model's performance across conditions, rather than collapsing everything into a single overloaded chart.

Conclusion

Focus on the priority areas first, then systematically work through the others. With dedicated effort, you can significantly strengthen your skills and project quality.

Assessment Date: 2026-05-18

This report provides developmental feedback to support your learning and growth.